

Reviewer Comments

Hearing Genomic Order: Windowed Sonification for Representing Mutation and Anomaly Patterns in Xylella fastidiosa Assemblies

Summary: The manuscript investigates whether window-based sonification can represent genomic sequence differences in *Xylella fastidiosa*. The authors analyze 50 RefSeq assemblies from five host-associated groups and test a 64-triplet pitch map and an amino-acid-class map. They compare these representations with standard sequence-composition features using leave-one-genome-out host-group classification. They also perform reference-guided anomaly detection using background substitutions, planted clustered substitutions, foreign-segment replacements and order shuffles. The strongest result is the detection of order-shuffled windows by sonification-derived features when triplet composition is preserved. The authors frame sonification as a representation tool rather than a replacement for alignment, variant calling or other standard genomic methods.

Major Issues – Must Fix:

1. The main comparison does not yet show that sonification adds information beyond standard order-sensitive sequence methods. The baseline contains base frequencies, GC proportion and 64 triplet frequencies so it is expected to fail when triplet counts are preserved. The authors should add an order-sensitive non-audio baseline, e.g. triplet-transition features or higher-order k-mer features such as 6-mers. They should also separate distribution features from interval/order features because the codon64 pitch histogram is mathematically equivalent to the 64-triplet frequency vector after relabeling. Add a compression-based baseline as well, such as normalized compression distance, which captures sequential structure without a biologically motivated mapping and gives a second, independent order-sensitive comparison. If the sonification features do not beat these order-sensitive baselines, reframe the result as evidence that order-sensitive features in general help on this anomaly class, not that sonification specifically is responsible.
2. The manuscript calls the analyzed three-base units “codons”. However, the Methods show that genomic FASTA sequences are divided directly into triplets without gene annotation, strand information or reading-frame information. Outside correctly framed coding regions these are nucleotide triplets rather than biological codons. The authors should correct this terminology or restrict the amino-acid analysis to properly annotated coding sequences.
3. You frame the work as sonification and build the title around hearing genomic order, but nothing in the evaluation is ever heard. The features are summary statistics of a codon-to-integer mapping, and you would get identical numbers by mapping the 64 codons to any 64 distinct values without sound at all. To keep the sonification framing, add a listener study showing that people can distinguish anomaly-containing tracks from background tracks above chance. If you cannot, reframe the contribution as a deterministic order-sensitive sequence encoding and remove the perceptual language (hearing, audibility, pitch-contour sensitivity) from the claims you test. The Discussion’s note that listener studies would be needed does not cover this, since the whole paper rests on sonification as the representation. The encoding and the anomaly benchmark are a real contribution either way. What needs to change is the claim that this is about hearing.
4. The FASTA preprocessing is not described clearly enough for a study based on sequence order. The authors state that headers are removed and only A, C, G and T are retained but they do not explain how separate contigs or ambiguous bases such as N are handled. The authors should

state the exact procedure and confirm that contig joining, or removal of ambiguous bases does not create artificial sequence order.

5. The dataset requires a clearer independence and selection check. Appendix Table A1 contains repeated strain names under different accessions, e.g. CFBP7970, CFBP8073, CFBP8173 and CFBP8075. If these are duplicate or near-identical assemblies, highly similar genomes could appear in both training and test sets. The authors should check genomic redundancy and also explain whether only 10 eligible genomes existed per host group or how the final 10 were selected.
6. Feature scaling is not described clearly. Pitch probabilities, entropy, mean MIDI pitch, standard deviations and interval measures have very different numerical ranges but Euclidean distance is used between feature vectors. It is also unclear whether classification features were scaled using training data only within each leave-one-genome-out fold. The authors should state the exact normalization and scaling procedure for both analyses.
7. The statistical support for the host-group result should be strengthened. Only 50 label permutations were used and $p = 0.0196$ is the minimum $1/51$ value under the common add-one calculation. This gives very limited p-value resolution. The authors should use more permutations and describe the permutation procedure clearly.
8. Reproducibility is limited because the scripts and outputs are reported as being stored in a local project workspace. External readers cannot reproduce the study from this location. The authors should provide an accessible repository with the scripts, software versions, accession metadata, parameters and representative outputs. The exact amino-acid-class to MIDI mapping, foreign-segment donor selection and score-to-MIDI transformation used for Figure 3 should also be documented.
9. Please expand the literature review and include at least 25–30 recent peer-reviewed primary research articles to strengthen the academic foundation of the manuscript.
10. Several citations are misrepresented, inaccurate, or over-extended to support claims their sources do not establish. Please audit every citation in the manuscript against the specific claim it is attached to, not merely the general topic.

Major Issues – Recommended:

1. The 40% host-group accuracy may partly reflect lineage structure rather than host-specific biology. *X. fastidiosa* has strong subspecies, phylogenetic and recombination structure as shown in comparative-genomic studies such as Vanhove et al. 2019, DOI: 10.1128/AEM.02972-18. At minimum, this possible confounding should be discussed clearly. A core-genome phylogeny or genomic-distance analysis would strengthen the interpretation if it is feasible at this stage.
2. The codon64 map places lexicographically ordered triplets on consecutive MIDI pitches 36–99. This means that pitch distances depend on the chosen numerical ordering. The Discussion already recognizes that the mapping is arbitrary. Testing one or more alternative or randomly permuted triplet-to-pitch mappings would help show whether the strong order-shuffle result depends on this specific encoding.

Minor Issues:

1. The Abstract reports codon64 performance for foreign-segment anomalies as AUC = 0.993 but does not mention that the composition baseline reaches AUC = 1.000 for the same condition. Please present this comparison more evenly, e.g. by noting that the composition baseline remained strongest for this composition-changing anomaly.
2. The anomaly-performance table in the Results needs a table number, title and clear caption, e.g. it currently begins directly with “Feature set / Anomaly / Mean AUC / 95% CI / Precision@4”

without a formal table label. Please ensure that all figures and tables are properly labeled, sequentially numbered, referenced within the text, and accompanied by clear and descriptive captions/titles.

3. Figure 3 needs enough information to stand alone. Please give the genome or accession, anomaly type and method used to select the example. The axes should also be labeled clearly.
4. The reference list should be thoroughly reviewed for consistency and standardization of formatting, capitalization, author names, journal titles, publication years, volume/issue numbers, page ranges, DOI information, and URLs. For example, Reference 15, DOI: 10.1186/s12859-021-04362-7, gives incorrect initials for Martin and Meagher. Reference 16, DOI: 10.1186/s13104-021-05685-7, also gives incorrect initials for Meagher. Reference 13 does not match the published 2022 GenBank author list.
5. Scientific names should be italicized consistently throughout the text and appendix, e.g. *Xylella fastidiosa*, *Prunus*, *Vitis*, *Olea* and *Coffea* should be italicized where appropriate. Please also correct clear wording errors such as “usecase” to “use case”.
6. Please ensure that all in-text citations follow consistent style, use superscript numerical formatting throughout the manuscript.

Manuscript Formatting Requirements

- **Font:** Times New Roman, 12 pt (headings should be bold).
- **Manuscript Length:** Maximum 20 pages with single line spacing, or 40 pages with double line spacing.
- **AI Usage:** You may use AI to offer suggestions on your work, but the writing must be your own. By submitting this revision, you confirm that this work was written entirely by the authors, without the use of AI. Use of AI will be grounds for rejection.